Chapter 2
Determinants

2.1 & 2.2

2.1 The Determinant of A Matrix

Objectives

- Determinant
- Minor and Cofactor
- Cofactor expansion

Determinants

- Recall that if A is nonsingular, then
  - A^{-1} exists,
  - $Ax = 0$ has only the **trivial (zero) solution** $x = 0$,
  - A is row equivalent to the identity I.
- The determinant of a **square matrix A** is a **scalar, denoted as $|A|$** (commonly computed using so-called expansion by minors).
  - $|A|$ is nonzero iff A has an inverse.
  - $|A| = 0$ iff A is singular

→ scalar / R

$\det(A)$ or $|A| \in R$

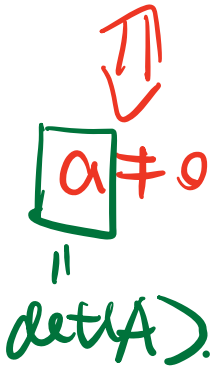
$$\begin{cases} \det(A) \neq 0 \iff A \text{ is nonsingular.} \\ \det(A) = 0 \iff A \text{ is singular.} \end{cases}$$

Logic of this section

- How to define determinant
  - Defined recursively from 1 by 1 and 2 by 2 matrix and the **minors** and **cofactors**
- How to compute determinant
  - What is minor
  - What is cofactor
  - Cofactor expansion

1×1 matrices

- A = $[a]$, then A will have an inverse $A^{-1} = [a^{-1}] = [1/a]$ if and only if $a \neq 0$.
- Thus, if we define $\det(A) = a$, then A will be nonsingular if and only if $\det(A) \neq 0$.



$A = [3]$

$\det(A) = 3.$

2×2 matrices

- Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, A is nonsingular iff A is row equivalent to I.
- Suppose $a_{11} \neq 0$, then we can test whether A is row equivalent to I by elementary row operations:

$$A \approx \begin{bmatrix} a_{11} & a_{12} \\ a_{11}a_{21} & a_{11}a_{22} \end{bmatrix} \approx \begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{11}a_{22} - a_{12}a_{21} \end{bmatrix}$$

- It will be row equivalent to I if and only if

$$a_{11}a_{22} - a_{12}a_{21} \neq 0 \quad (1)$$

- How about if a_{11}=0?

$$A \xrightarrow{R1 \leftrightarrow R2} \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix} = \begin{bmatrix} a_{21} & a_{22} \\ 0 & a_{12} \end{bmatrix}$$

- It will be row equivalent to I if and only if $a_{12}a_{21} \neq 0$ (Equation1 when a_{11}=0)

- Thus, we define $\det(A) = a_{11}a_{22} - a_{12}a_{21}$, A is nonsingular iff $\det(A) \neq 0$

$n \times n$ A is nonsingular $\Leftrightarrow$ A is row equivalent to I.

$\Updownarrow$ S.T.F $\nearrow$

(1) $a_{11} \neq 0$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \xrightarrow{R_2 - \frac{a_{21}}{a_{11}} R_1} \begin{bmatrix} a_{11} \neq 0 & a_{12} \\ 0 & a_{22} - \frac{a_{12}a_{21}}{a_{11}} \end{bmatrix}$$

$\neq 0$

$a_{22} - \frac{a_{12}a_{21}}{a_{11}} \neq 0 \Leftrightarrow A$ is nonsingular.

$\Updownarrow$ $\searrow$

$a_{11}a_{22} - a_{12}a_{21} \neq 0$

(2) $a_{11} = 0$

$$A = \begin{bmatrix} 0 & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} a_{21} & a_{22} \\ 0 & a_{12} \end{bmatrix}$$

$a_{21} \neq 0$ & $a_{12} \neq 0$ $\Leftrightarrow$ $a_{12}a_{21} \neq 0$

$\Leftrightarrow$ $a_{11}a_{22} - a_{12}a_{21} \neq 0 \Leftrightarrow A$ is nonsingular

$=$

0

(1) & (2)

$a_{11}a_{22} - a_{12}a_{21} \neq 0 \iff A$ is non-singular.

$\|$

det(A)

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Eg. $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

$|A| = 1 \times 2 - 1 \times 2 -$

Notation

The **determinant** of a square matrix A is denoted by $|A|$.
The notation $\det(A)$ is also used for the determinant of A.

Example

If $A = \begin{bmatrix} 2 & 4 \\ -3 & 1 \end{bmatrix}$, then $\det(A) = \begin{vmatrix} 2 & 4 \\ -3 & 1 \end{vmatrix} = 2 \times 1 - (-3) \times 4 = 14$

Ch2.2

$\det(A) \neq 0 \Leftrightarrow A$ is nonsingular

$\det(A) = 0 \Leftrightarrow A$ is singular.

$n \times n$ matrices

- The determinant of an $n \times n$ matrix A is defined recursively through **cofactors**

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{i=1}^{n} a_{ij}A_{ij}$$

- where $A_{ij} = (-1)^{i+j}\det(M_{ij})$ is the **cofactor** of element a_{ij},
- M_{ij} is an $(n-1) \times (n-1)$ matrix, obtained by delete the ith row and jth column of A.
- $\det(M_{ij})$ is called the **minor** of a_{ij} of A.
- The cofactor expansion transform the determinant of an $n \times n$ matrix into the summation of n determinants of $(n-1) \times (n-1)$ matrices
- The cofactor expansion can be operated on any row or any column.
- We usually choose the row or column which has many zeros.

$$|A| = \sum_{j=1}^{n} a_{ij} A_{ij} \qquad A: n \times n$$

(1) a_{ij} : entry in the i-th row and j-th column. $\in R$

(2) A_{ij} : cofactor of a_{ij} $\in R$

$$A_{ij} = (-1)^{i+j} \underbrace{\det(M_{ij})}_{\text{minor}} \in R$$

(3) M_{ij} : $(n-1) \times (n-1)$ deleting i-th row j-th column from A.

Ex. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3\times 3}$

$M_{12} = \begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix}_{2\times 2}$

$M_{23} = \begin{bmatrix} 1 & 2 \\ 7 & 8 \end{bmatrix}_{2\times 2}$

$$|A| = \sum_{j=1}^{n} a_{ij} A_{ij}$$

$$A_{ij} = (-1)^{i+j} |M_{ij}|$$

$$= a_{i1} A_{i1} + a_{i2} A_{i2} + \cdots + a_{in} A_{in}$$

cofactor expansion along the i-th row

$$= a_{i1}(-1)^{i+1} |M_{i1}| + \cdots + a_{in}(-1)^{i+n} |M_{in}|$$

$$|A| = \sum_{i=1}^{n} a_{ij} A_{ij}$$

$$= a_{1j} A_{1j} + a_{2j} A_{2j} + \cdots + a_{nj} A_{nj}$$

cofactor expansion along the j-th column

Eg. $|A| = \begin{vmatrix} 0 & 1 & 2 \\ 0 & 3 & 4 \\ 5 & 6 & 7 \end{vmatrix} = a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13}$

($a_{11} = 0$)

$$= a_{12} A_{12} + a_{13} A_{13}$$

$$= a_{11} A_{11} + a_{21} A_{21} + a_{31} A_{31} = 5 A_{31}$$

($a_{11} = 0$, $a_{21} = 0$, $a_{31} = 5$)

Theorem : Cofactor expansions

Definition 1st row cofactor expansion

The **determinant of a square matrix** is the sum of the products of the elements of the first row and their cofactors.

$$\text{If } A \text{ is } 3\times 3, |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$$\text{If } A \text{ is } 4\times 4, |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + a_{14}A_{14}$$

$$\vdots$$

$$\text{If } A \text{ is } n\times n, |A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + \cdots + a_{1n}A_{1n}$$

These equations are called **cofactor expansions** of $|A|$.

- Here A_{ij} is denoted for the cofactor, usually, my habit is use C_{ij} for cofactor ofa_{ij}

3×3. $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$A_{ij} = (-1)^{i+j} |M_{ij}|$

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

$$= a_{11}(-1)^{1+1}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1)^{1+2}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

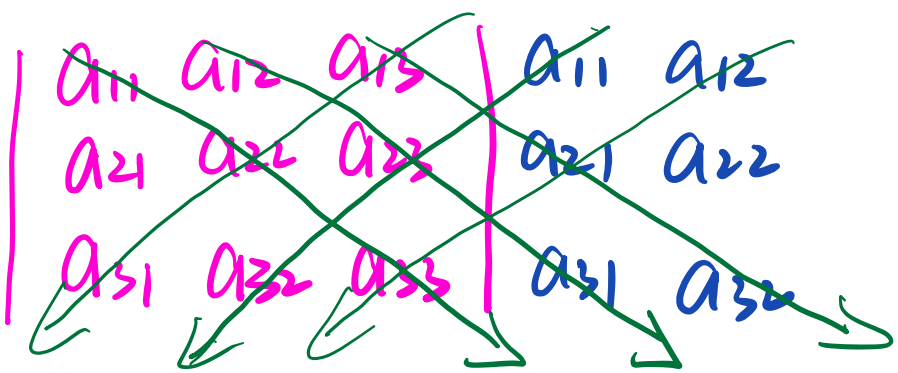
$$+ a_{13}(-1)^{1+3}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31})$$

$$+ a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$

$$- a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$



If $n \geq 4$. cofactor expansion

The above method is not applicable.

Example

Determine the minors and cofactors of the elements a_{11} and a_{32} of the following matrix A.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{bmatrix}$$

Solution

$$\text{Minor of } a_{11}: |M_{11}| = \begin{vmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} = (-1\times 1) - (2\times(-2)) = 3$$

$$\text{Cofactor of } a_{11}: A_{11} = (-1)^{1+1}|M_{11}| = (-1)^2(3) = 3$$

$$\text{Minor of } a_{32}: |M_{32}| = \begin{vmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 4 & 2 \end{vmatrix} = (1\times 2) - (3\times 4) = -10$$

$$\text{Cofactor of } a_{32}: A_{32} = (-1)^{3+2}|M_{32}| = (-1)^5(-10) = 10$$

Examples

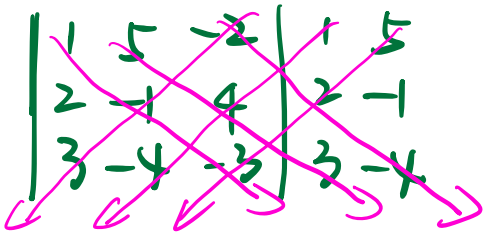
- Find the determinants of the following matrices:

$|A| = 3 - 24 = -21$ $\quad |B| = \det(B) = 12 - 12 = 0$

$$A = \begin{bmatrix} 1 & 4 \\ 6 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & -2 \\ -6 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 5 & -2 \\ 2 & -1 & 4 \\ 3 & -4 & -3 \end{bmatrix} \qquad D = \begin{bmatrix} a & 0 & 0 \\ 5 & b & 0 \\ 6 & 7 & c \end{bmatrix}$$

$|C| =$



$= 3 + 60 + 16 + 30 + 16 - 6$

$= 119$

$|D| = a(-1)^{1+1} \begin{vmatrix} b & 0 \\ 7 & c \end{vmatrix}$

$= abc$

Examples

Evaluate the determinant of the following matrix *A*.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{matrix} 1 & 2 \\ 3 & 0 \\ 4 & 2 \end{matrix}$$

$$|A| = 3 \times (-1)^{2+1} \begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} + 1 \times (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix}$$

$$= -3(2+2) - (2-8)$$

$$= -12 + 6 = -6$$

Examples

Evaluate the determinant of the following matrix A.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

Solution

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$
$$= a_{11}(-1)^{1+1} \det M_{11} + a_{12}(-1)^{1+2} \det M_{12}$$
$$+ a_{13}(-1)^{1+3} \det M_{13}$$
$$= 1\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} \text{-2} \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 3 & 0 \\ 4 & 1 \end{vmatrix}$$
$$= [(0 \times 1) - (1 \times 2)] - 2[(3 \times 1) - (1 \times 4)] - [(3 \times 2) - (0 \times 4)]$$
$$= -2 + 2 - 6$$
$$= -6$$

Example: 4x4 determinants

Find |A|, where $A = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & -1 & -1 & 2 \\ 0 & -3 & 1 & 0 \\ -3 & 3 & 1 & 0 \end{bmatrix}$

- Cofactor expansion along the first row: need to calculate 3 more 3x3 determinants (each will expand into 3 more 2x2 determinants…)
- The 4th column of A has two 0's. Can we calculate determinants by cofactor expansion along this column instead? Will we still get the same answer?

$|A| = a_{14}A_{14} + a_{24}A_{24}$

$= 1\times(-1)^{1+4}\begin{vmatrix} 2 & -1 & -1 \\ 0 & -3 & 1 \\ -3 & 3 & 1 \end{vmatrix} + 2\times(-1)^{2+4}\begin{vmatrix} 1 & 0 & -1 \\ 0 & -3 & 1 \\ -3 & 3 & 1 \end{vmatrix}$

$$= -1 \times \left(2 \times (-1)^{1+1} \begin{vmatrix} -3 & 1 \\ 3 & 1 \end{vmatrix} + (-3) \times (-1)^{3+1} \begin{vmatrix} -1 & -1 \\ -3 & 1 \end{vmatrix} \right)$$

$$+ 2 \times \left(1 \times (-1)^{1+1} \begin{vmatrix} -3 & 1 \\ 3 & 1 \end{vmatrix} + (-1) \times (-1)^{1+3} \begin{vmatrix} 0 & -3 \\ -3 & 3 \end{vmatrix} \right)$$

$$= -\left(2 \times (-6) - 3 \times (-4) \right)$$

$$+ 2 \left(-6 - (-9) \right)$$

$$= -(-12 + 12) + 2(-6 + 9)$$

$$= 0 + 6 = 6$$

Theorem: Cofactor expansion

The determinant of a square matrix is the sum of the products of the elements of any row or column and their cofactors.

- i^{th} row expansion: $|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$
- j^{th} column expansion: $|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}$

Example Find the determinant of the following matrix.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

$$B = \begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

$$|B| = 2 \times (-1)^{4+1} \begin{vmatrix} 2 & 3 & 0 \\ 4 & 5 & 0 \\ 1 & 0 & 3 \end{vmatrix}$$

$$= -2 \times 3 \times (-1)^{3+3} \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$$

$= -6\,(10-12) = 12$

Theorem: Cofactor expansion

The determinant of a square matrix is the sum of the products of the elements of any row or column and their cofactors.

- i^{th} row expansion: $|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$
- j^{th} column expansion: $|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}$

Example Find the determinant of the following matrix.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

Solution

$$\begin{aligned} |A| &= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} \\ &= -3\begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & -1 \\ 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} \\ &= -3[(2\times 1) - (-1\times 2)] + 0[(1\times 1) - (-1\times 4)] - 1[(1\times 2) - (2\times 4)] \\ &= -12 + 0 + 6 = -6 \end{aligned}$$

Example

Solve the following equation for the variable x.

$$\begin{vmatrix} x & x+1 \\ -1 & x-2 \end{vmatrix} = 7$$

$x(x-2) - (-1)(x+1) = 7$

$x^2 - 2x + x + 1 = 7$

$x^2 - x - 6 = 0$

$(x+2)(x-3) = 0$

$x = -2$ or $x = 3$

Example

Solve the following equation for the variable x.

$$\begin{vmatrix} x & x+1 \\ -1 & x-2 \end{vmatrix} = 7$$

Solution Expand the determinant to get the equation

$$x(x-2)-(x+1)(-1)=7$$

Proceed to simplify this equation and solve for *x*.

$$\begin{aligned} x^2-2x+x+1&=7 \\ x^2-x-6&=0 \\ (x+2)(x-3)&=0 \\ x&=-2 \text{ or } 3 \end{aligned}$$

There are two solutions to this equation, $x = -2$ or 3.

Example

Evaluate the determinant of the following 4×4 matrix.

$$A = \begin{bmatrix} 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 7 & -2 & 3 & 5 \\ 0 & 1 & 0 & -3 \end{bmatrix}$$

$$|A| = 3 \times (-1)^{3+3} \begin{vmatrix} 2 & 1 & 4 \\ 0 & -1 & 2 \\ 0 & 1 & -3 \end{vmatrix}$$

$$= 3 \times 2 \times (-1)^{1+1} \begin{vmatrix} -1 & 2 \\ 1 & -3 \end{vmatrix}$$

$$= 6\,(3 - 2) = 6$$

Example

Evaluate the determinant of the following 4×4 matrix.

$$\begin{bmatrix} 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 7 & -2 & 3 & 5 \\ 0 & 1 & 0 & -3 \end{bmatrix}$$

Solution 3rd column expansion:

$$|A| = a_{13}A_{13} + a_{23}A_{23} + a_{33}A_{33} + a_{43}A_{43}$$

$$= 0A_{13} + 0A_{23} + 3A_{33} + 0A_{43}$$

$$= 3\begin{vmatrix} 2 & 1 & 4 \\ 0 & -1 & 2 \\ 0 & 1 & -3 \end{vmatrix}$$

$$= 3 \times 2\begin{vmatrix} -1 & 2 \\ 1 & -3 \end{vmatrix} = 6 \times (3 - 2) = 6$$

upper & lower.

Determinants of Triangular Matrices

Theorem 1 If A is an $n \times n$ **triangular** matrix, then the determinant of A equals the product of the diagonal elements of A.

Idea: expansion along rows or columns with lots of 0's.

Example: $\begin{vmatrix} 2 & 0 & 0 \\ 10 & 3 & 0 \\ 100 & 1000 & 4 \end{vmatrix} = 2 \begin{vmatrix} 3 & 0 \\ 1000 & 4 \end{vmatrix} = 2 \times 3 \times 4$

- Directly apply the cofactor expansion
- By induction: suppose the statement holds for $(n-1) \times (n-1)$ matrices, that is, the statement is true for its **minors.**

Case I: lower triangular.

$$|A| = \begin{vmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

$$= a_{11}(-1)^{1+1} \begin{vmatrix} a_{22} & 0 & \cdots & 0 \\ a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

$(-1)^{1+1} = 1$

$$= a_{11}a_{22}(-1)^{1+1} \begin{vmatrix} a_{33} & 0 & \cdots & 0 \\ a_{43} & a_{44} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n3} & a_{n4} & \cdots & a_{nn} \end{vmatrix}$$

$$= a_{11}a_{22}a_{33}\cdots a_{nn}$$

Case II Upper Triangular.

$$|A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix}$$

$$= a_{11}(-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & - & a_{nn} \end{vmatrix}$$

$$= a_{11} a_{22} (-1)^{1+1} \begin{vmatrix} a_{33} & a_{34} & \cdots & a_{3n} \\ 0 & a_{44} & \cdots & a_{4n} \\ 0 & 0 & \ddots & a_{nn} \end{vmatrix}$$

$$\vdots$$

$$= a_{11} a_{22} a_{33} \cdots a_{nn}$$

Q: $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 1 \times 0 \times 2 = 0$

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diagonal matrix is both upper & lower triangular.

Properties of Determinants

Theorem 2

- If A has a row or column consisting entirely zeros, then $\det(A) = 0$.
- If A has two identical rows or columns, then $\det(A) = 0$.

Expansion according to the ith row

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

- Directly apply the cofactor expansion
- By induction: suppose the statement holds for $(n-1) \times (n-1)$ matrices, that is, the statement is true for its minors.

(2) Mathematical Induction method

To show statement p is true.

① When $n=1$, p is true.

② When $n=2$, p is true.

③ Assume $n=k$. p is true

④ Verify. when $n=k+1$, p is still true based on the assumption at $n=k$.

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(1) When $n=2$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{11} & a_{12} \end{vmatrix} = a_{11}a_{12} - a_{11}a_{12} = 0$$

(2) When $n=3$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{21} & a_{22} & a_{23} \end{vmatrix}$$

$$= a_{11}(-1)^{1+1}\begin{vmatrix} a_{22} & a_{23} \\ a_{22} & a_{23} \end{vmatrix} + a_{12}(-1)^{1+2}\begin{vmatrix} a_{21} & a_{23} \\ a_{21} & a_{23} \end{vmatrix}$$

$$+ a_{13}(-1)^{1+3}\begin{vmatrix} a_{21} & a_{22} \\ a_{21} & a_{22} \end{vmatrix} = 0 + 0 + 0 = 0$$

③ Assume. $n=k$.

if A has two identical rows.

$\det(A)=0$

④ When $n=k+1$

Assume $R_i = R_j$

$$|A| = a_{m1}A_{m1} + \cdots + a_{mn}A_{mn}$$

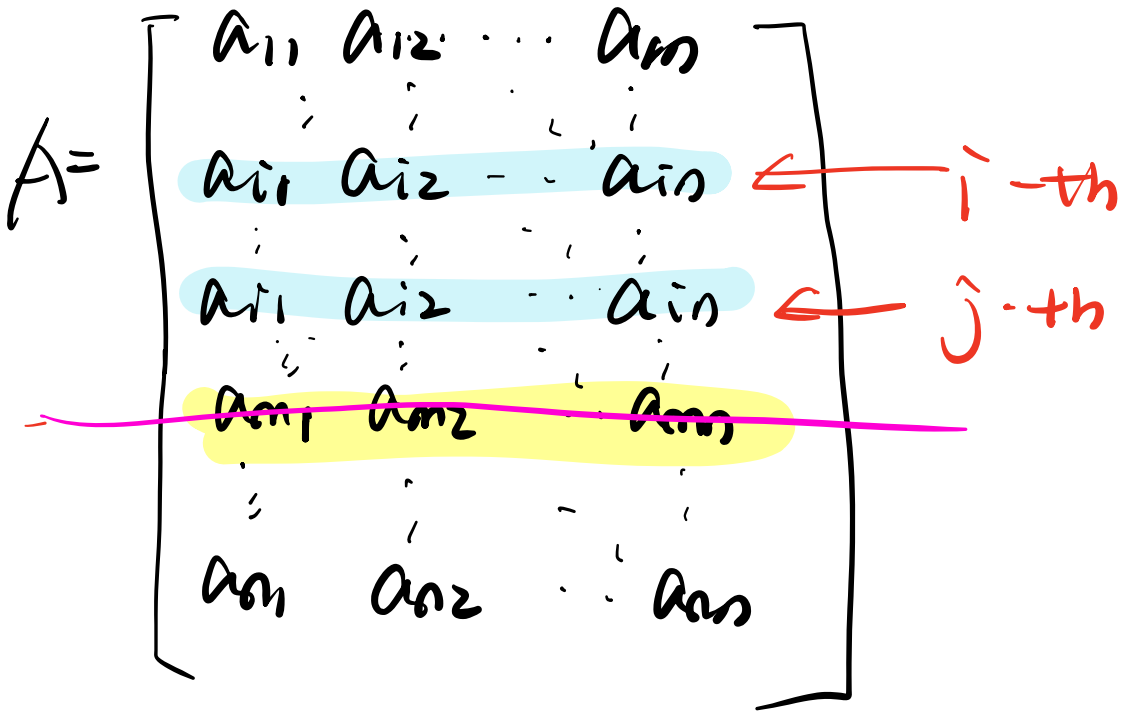
cofactor expansion along the m-th row $(m \neq i \ \& \ m \neq j)$

$$= a_{m1}(-1)^{m+1}\underbrace{|M_{m1}|}_{=0} + \cdots$$

$$+ a_{mn}(-1)^{m+n}\underbrace{|M_{mn}|}_{=0}$$

$$= 0 + 0 + \cdots + 0$$

$$= 0$$



M_{n1} ⋯ M_{nn} are $k \times k$ matrices that have two identical rows.

⇓

$$|M_{n1}| = \cdots = |M_{nn}| = 0$$